



IDX G9 PHYSICS H STUDY GUIDE

ISSUE S2 FINALS

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This guide is a combined guide. Due to the final range including content from before the monthlies, we have included content from the previous issue in this guide for your convenience.

Lesson 6.1: Work Done by a Constant Force

Definition of Work

Work is defined as the product of the force applied to an object and the displacement of the object in the direction of the force.

- Symbol: W
- Unit: Joule (J), where $1 \text{ J} = 1 \text{ N} \cdot \text{m}$
- Formula:

$$W = Fd \cos \theta$$

Here, F is the magnitude of the force, d is the displacement, and θ is the angle between the force and displacement vectors.

Special Cases

- When $\theta = 0^\circ$: $W = Fd$ (maximum positive work).
- When $\theta = 180^\circ$: $W = -Fd$ (negative work, e.g., friction).
- When $\theta = 90^\circ$: $W = 0$ (no work done, e.g., centripetal force).

Example 1: Sailor Pulling a Boat

A sailor pulls a boat with a force of 255 N at an angle of 25.0° over a distance of 0.30 m. The work done is:

$$W = 255 \times 0.30 \times \cos 25.0^\circ = 69 \text{ J}$$

Example 2: Crate on a Rough Floor

A 50-kg crate is pulled 40 m by a 100 N force at 37° on a rough floor ($\mu_k = 0.1$):

$$W_{\text{pull}} = 100 \times 40 \times \cos 37^\circ \approx 3195 \text{ J}$$

$$W_{\text{friction}} = -0.1 \times (50 \times 9.8 - 100 \sin 37^\circ) \times 40 \approx -1720 \text{ J}$$

Net work: $W_{\text{net}} = 3195 - 1720 = 1475 \text{ J}$.

Lesson 6.2: Work Done by a Varying Force

Graphical Method

For a varying force, work is calculated as the area under the force-displacement ($F - d$) graph:

- Positive area: Work done by the force.
- Negative area: Work done against the force.

Example 3: Spring Force

A spring exerts a linearly increasing force from 0 N to 20 N over 1.50 m. The work done is:

$$W = \frac{1}{2} \times 20 \times 1.50 = 15.0 \text{ J}$$

Example 4: Trapezoidal Force

A force varies from 150 N to 250 N over 25.0 m. The work done is:

$$W = \frac{1}{2}(150 + 250) \times 25 = 5000 \text{ J}$$

Example 5: Piecewise Force

A force varies as follows:

- From $x = 0$ to $x = 10$ m: $W = 2800 \text{ J}$.
- From $x = 10$ m to $x = 15$ m: $W = -700 \text{ J}$.
- Total work: $2800 - 700 = 2100 \text{ J}$.

Additional Examples

Example 6: Hiker Carrying a Backpack

A hiker carries a 15.0 kg backpack up a 10.0 m hill at constant velocity:

$$W_{\text{hiker}} = 15.0 \times 9.8 \times 10.0 = 1470 \text{ J}$$

$$W_{\text{gravity}} = -1470 \text{ J}$$

Net work: 0 J.

Example 7: Arrow Fired from a Bow

An 88 g arrow is fired with a 110 N force over 78 cm:

$$v = \sqrt{\frac{2 \times 110 \times 0.78}{0.088}} \approx 44 \text{ m/s}$$

6-3 Kinetic Energy and Work-Energy Principle

1. Kinetic energy (Translational kinetic energy)
 - a. Energy
 - i. Traditional way: the ability to do work.
 - ii. The ability of an object to produce a change in itself or the world around
 - b. Translational KE
 - i. Definition – the energy of motion
 - ii. Symbol – KE
 - iii. Unit – J (Joule)
 - iv. Formula $\frac{1}{2}mv^2$
2. Work-energy principle
 - a. Formula: $W_{net} = \Delta KE = KE_f - KE_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$
 - b. Statement: The net work done on an object is equal to the change in the object's kinetic energy
 - c. The work-energy principle is valid only if W is the net work done on the object

6-4 Potential Energy

1. Potential energy: the energy associated with forces that depend on the position or configuration of an object relative to the surroundings.
2. Forms of PE:
 - a. Gravitational potential energy
 - i. Definition: The stored energy in a system resulting from the gravitational force between Earth and the object.
 - ii. Symbol: PE_{grav} or PE_g
 - iii. Formula: $PE_g = mgh$, $\Delta PE_g = mg\Delta h$

1. m = mass
 2. g = acceleration due to gravity
 3. h = height above reference level
- iv. Work done by external force to move the object from point 1 to point 2 (with no acceleration) = change in potential energy
1. $W_{ext} = \Delta PE$
- v. PE belongs to a system and not a single object
1. PE is associated with a force
 2. Force on one object is always exerted by some other object
- b. Elastic Potential Energy
- i. Definition: the stored energy in an object as a result of its change in shape
 - ii. E.g. rubber bands, springs
 - iii. Hooke's Law
 1. Statement: The spring force is directly proportional to the displacement of the spring and opposite in the direction of the spring
 2. Formula: $F_s = -kx$, $PE_e = \frac{1}{2}kx^2$
 - a. k : spring stiffness constant
 - b. x : distance stretched/compressed

6.5 Conservative and Nonconservative Forces

- Conservative – energy is not lost, $W_{net} = 0$
- Nonconservative – energy is lost, forces that removes energy from the system

6.6 & 6.7 Mechanical Energy and its Conservation

- Mechanical Energy: Sum of kinetic and potential energies in the system
- $E = KE + PE$
- Principle of Conservation of Mechanical Energy: When only conservative forces act on a system, its total mechanical energy is conserved
- $E_i = KE_i + PE_i = E_f = KE_f + PE_f$
- $KE_i + GPE_i + EPE_i = KE_f + GPE_f + EPE_f$

6.8 & 6.9 Law of Conservation of Energy

The Law of Conservation of Energy

1. Transformation of energy
 - a. Energy can be transformed from one form to another
 - b. Work is done when energy is transferred from one object to another
2. Law of conservation of energy
 - a. The total energy is neither increased nor decreases in any process. Energy can be transferred from one form to another, and transferred from one object to another, but the total amount remains constant.
3. Energy conservation with dissipative forces
 - a. Dissipative force: the force that reduces the mechanical energy but not the total energy
 - i. E.g., friction
 - b. The thermal energy produced by a constant friction force is equal to the work done by friction.
 - c. Conservation of energy with gravity and friction

i. Statement: The initial mechanical energy of an object equals the final mechanical energy plus the energy transformed by friction into thermal energy.

ii. Formula: $E_i = E_f \rightarrow \frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f + F_{fr}d$

6.10 Power

Power

1. Definition:

- a. The rate at which work is done
- b. Average power equals the work done divided by the time to do it
- c. The rate at which energy is transformed

2. Formula: $\bar{P} = \frac{W}{t} = \frac{\Delta E}{t}$

3. SI Unit: Watt (W)

- a. $1\text{ W} = 1\text{ J/s}$
- b. $1\text{ kW} = 1000\text{ W}$
- c. $1\text{ hp (horsepower)} = 746\text{ W}$

4. Force and displacement should be in the same direction

Efficiency

1. Definition: the ratio of the useful power output of the engine to the power input

2. Formula: $e = \frac{P_{out}}{P_{in}}$

3. Efficiency is always less than 1.0

- a. No engine can create energy
- b. No engine can even transform energy from one form to another without some energy going to friction, thermal energy and other nonuseful forms of energy.

7.1 Momentum and Its Relation to Force

1. Momentum

- a. Definition: the product of the object's mass and its velocity
- b. Formula: $\vec{p} = m\vec{v}$
- c. Vector: direction same as its velocity
- d. Unit: kg*m/s
- e. Change in momentum:
 - i. $\Delta\vec{p} = \vec{p}_f - \vec{p}_i = m\vec{v}_f - m\vec{v}_i = m\Delta\vec{v}$
- f. You must define a positive direction when solving problems involving impulse and momentum.

7.2 Conservation of Momentum

1. Conservation of Momentum:

- a. Under certain circumstances, momentum is a conserved quantity: no external force is exerted on the system.
- b. Momentum of the system is conserved: The sum of the momenta of the colliding objects is the same before and after the collision.
- c. Formula: $\vec{p}_{Ai} + \vec{p}_{Bi} = \vec{p}_{Af} + \vec{p}_{Bf}$
- d. The law of conservation of momentum: The momentum of an isolated system of objects remains constant.
 - i. Isolated system: the system in which the only (significant) forces are those between the objects in the system.

2. Recoil problem

- a. Examples: rocket propulsion, gun recoil
 - i. For rocket propulsion, consider the rocket and fuel as an isolated system with no external forces.

- ii. In the reference frame of the rocket, the total momentum of rocket plus fuel is zero.
- b. After explosion
 - i. Momenta of the two objects: equal in magnitude, opposite in direction
 - ii. Velocities of the two objects: $\vec{v}_{Af} = -\frac{m_B}{m_A}\vec{v}_{Bf}$
 - 1. The velocities depend on their relative masses
 - 2. Less massive object $\rightarrow$ greater velocity

7.3 Impulse-Momentum Theorem

1. Impulse:
 - a. Definition: the product of the force on an object and the time interval over which it acts.
 - b. Formula: $\vec{I} = \vec{F}\Delta t$
 - c. Vector: direction is the same as the force
 - d. Unit: Newton-second (N*s)
 - e. When force is constant: $I=Ft$
 - f. When force is not constant: Impulse is the area under a F-t graph
2. Newton's Second Law of motion in terms of momentum: $\sum \vec{F} = \frac{\Delta \vec{p}}{\Delta t}$
 - a. The rate of change of momentum of an object is equal to the net force applied to it.
3. Impulse-Momentum Theorem: $\vec{I} = \Delta \vec{p} \rightarrow \vec{F}\Delta t = \vec{p}_f - \vec{p}_i$
 - a. The impulse on an object is equal to the object's final momentum minus the object's initial momentum.

Chapter 7: Linear Momentum (Lessons 7.4 to 7.6)

7-4 Conservation of Energy and Momentum in Collisions

Three Kinds of Collisions

- **Elastic Collision:** Total kinetic energy is conserved.
- **Inelastic Collision:** Total kinetic energy decreases after collision.
- **Superelastic/Explosive Collision:** Total kinetic energy increases after collision (e.g., explosions).

7-5 Elastic Collisions in One Dimension

Definition and Characteristics

- **Definition:** A collision where total kinetic energy is conserved.
- **Characteristics:**
 - Conservation of momentum:

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

- Conservation of kinetic energy:

$$\frac{1}{2}m_A v_{Ai}^2 + \frac{1}{2}m_B v_{Bi}^2 = \frac{1}{2}m_A v_{Af}^2 + \frac{1}{2}m_B v_{Bf}^2$$

Head-on Elastic Collision (1D)

For objects colliding head-on:

$$v_{Ai} - v_{Bi} = -(v_{Af} - v_{Bf})$$

The relative speed after the collision has the same magnitude but opposite direction as before the collision.

Example: Elastic Collision of Equal Masses

If $m_A = m_B$, the velocities exchange:

$$v_{Af} = 0, \quad v_{Bf} = v_{Ai}$$

7-6 Inelastic Collisions

Definition and Characteristics

- **Definition:** A collision where total kinetic energy decreases.
- **Characteristics:**
 - Momentum is conserved:

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

- Kinetic energy is not conserved; some is converted to other forms (e.g., thermal energy).

Completely Inelastic Collision

- Objects stick together after collision.
- Momentum conservation:

$$m_A v_{Ai} + m_B v_{Bi} = (m_A + m_B) v_f$$

- Maximum kinetic energy is lost.

Example: Ballistic Pendulum

- Part (1): Conservation of momentum to find velocity after collision:

$$v' = \frac{mv}{m + M}$$

- Part (2): Conservation of energy to find maximum height h :

$$v' = \sqrt{2gh}$$

- Combine to find initial projectile speed:

$$v = \frac{m + M}{m} \sqrt{2gh}$$

Energy Loss in Inelastic Collisions

Calculate the difference in kinetic energy before and after the collision:

$$\Delta KE = KE_f - KE_i$$

For completely inelastic collisions, this loss is maximized.